

STEI, Romania

WMO: 151600

Lat: 46.5281N

Lon: 22.4664E

Elev: 916

StdP: 14.22

Time Zone: 2.00 (EUE)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.7	17.5	8.0	8.6	16.0	12.1	10.6	20.7	18.3	34.4	16.1	34.7	3.7	180	0.475

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.2	89.9	69.4	86.3	69.1	83.2	67.9	72.0	83.6	70.6	82.1	69.3	80.6	5.5	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.4	108.1	76.4	67.0	102.7	75.3	65.6	97.6	73.9	36.2	83.7	35.1	82.1	33.9	81.1	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.3	13.2	11.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.9	31.5	37.4	43.2	51.5	60.2	66.7	68.7	69.7	60.6	52.2	44.8	35.3
	DBStd	14.84	8.53	7.91	7.68	8.52	5.76	5.82	5.19	6.07	7.56	7.91	8.36	7.62
	HDD50	1987	573	356	236	86	5	0	0	0	3	72	199	457
	HDD65	5255	1037	773	677	408	165	49	24	24	174	397	606	921
	CDD50	2683	1	3	25	129	322	502	581	611	323	141	43	2
	CDD65	475	0	0	0	2	17	102	140	169	43	2	0	0
	CDH74	4796	0	0	4	55	229	938	1402	1706	418	44	0	0
	CDH80	1545	0	0	0	2	21	253	459	689	120	1	0	0
(14) Wind	WSAvg	5.0	5.0	5.5	5.1	5.2	5.2	4.7	4.9	4.7	4.8	5.0	5.3	5.2
(15) Precipitation	PrecAvg	38.3	2.7	2.2	2.6	3.1	3.6	4.4	4.3	2.9	3.5	3.1	3.0	3.2
	PrecMax	50.4	4.8	5.6	8.0	5.2	6.8	8.4	8.1	7.1	8.8	7.0	5.6	6.8
	PrecMin	27.4	1.1	0.4	0.8	0.8	1.5	1.4	1.9	0.4	0.6	0.3	0.2	0.3
	PrecStd	6.1	0.9	1.2	1.7	1.2	1.4	1.6	1.9	1.6	1.9	1.7	1.2	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.6	60.9	71.5	79.1	82.7	89.3	92.3	94.7	89.4	78.2	68.7	57.3
		MCWB	46.7	49.7	57.0	60.4	64.1	71.5	69.8	69.7	67.0	60.5	55.3	48.2
	2%	DB	48.8	56.3	64.7	75.0	79.5	86.0	88.4	90.9	82.6	74.6	64.5	51.9
		MCWB	43.7	47.5	51.4	57.9	63.9	70.6	69.0	68.9	65.1	59.9	54.1	45.8
	5%	DB	45.7	52.3	59.7	70.9	76.6	82.8	85.0	86.9	79.0	70.3	60.7	48.6
		MCWB	41.9	45.7	49.4	56.1	62.8	68.8	68.5	68.8	64.0	58.6	52.2	43.7
	10%	DB	42.7	48.9	55.2	66.4	73.5	79.8	81.7	83.1	74.0	65.1	57.3	45.5
		MCWB	40.0	43.7	46.8	53.7	61.3	67.7	67.2	68.3	62.6	56.4	50.9	42.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.9	51.6	56.7	61.9	67.8	74.6	72.8	73.5	68.8	63.1	56.6	49.9
		MCDB	52.8	58.3	68.9	75.6	77.6	85.1	84.1	87.0	84.9	72.2	64.8	55.8
	2%	WB	45.1	48.3	53.1	59.4	65.7	72.0	71.3	71.8	66.9	60.9	55.0	47.0
		MCDB	48.0	54.0	62.7	72.7	76.0	82.3	82.7	84.5	80.9	71.5	62.6	51.7
	5%	WB	42.3	46.6	50.1	57.3	63.9	70.4	70.1	70.2	65.1	59.3	53.4	44.2
		MCDB	45.2	52.0	58.5	68.6	73.6	80.6	81.3	82.9	76.1	68.3	59.7	48.0
	10%	WB	39.8	44.3	47.8	55.3	62.2	68.6	68.8	69.0	63.6	57.2	51.5	42.3
		MCDB	42.3	48.2	53.9	64.8	71.1	78.1	79.3	81.2	72.8	64.7	57.0	45.5
(35) Mean Daily Temperature Range	5% DB	MDBR	10.9	13.6	17.4	19.7	20.1	20.6	22.0	22.2	19.2	18.7	13.7	9.9
		MCDBR	12.5	18.5	24.6	26.2	26.0	25.0	27.1	27.8	26.4	25.8	18.8	14.5
		MCWBR	8.1	11.0	13.2	12.0	12.6	11.3	11.2	10.4	10.8	12.9	10.1	9.1
	5% WB	MCDBR	11.6	15.8	22.3	23.8	22.8	23.2	23.6	24.7	23.3	23.1	16.7	13.7
		MCWBR	8.0	9.9	12.4	11.3	11.8	11.2	10.8	10.6	10.1	11.9	9.0	9.1
(40) Clear-Sky Solar Irradiance	taub		0.326	0.356	0.401	0.456	0.441	0.436	0.453	0.458	0.421	0.383	0.353	0.327
	taud		2.308	2.241	2.175	2.052	2.161	2.215	2.174	2.157	2.224	2.318	2.346	2.346
	Ebn at Noon		238	253	258	253	263	265	258	250	248	241	226	223
	Edh at Noon		25	32	40	49	46	44	45	44	38	30	24	22
(44) All-Sky Solar Radiation	RadAvg		380	633	1000	1380	1665	1839	1800	1661	1180	808	462	299
	RadStd		53	83	145	141	159	154	159	159	134	96	63	40

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (63 neighbors)	+1.26	N/A	N/A	N/A	+1.78	N/A	N/A	-234	-355	+207	+104

Nomenclature: See separate page